

Establishing reference ranges for ambulatory electrocardiography parameters: meta-analysis

Curtis B Williams,¹ Jason G Andrade,² Nathaniel M Hawkins,² Christopher Cheung,² Andrew Krahn,² Zachary W Laksman,² Matthew T Bennett,² Brett Heilbron,¹ Shanta Chakrabarti,² John A Yeung-Lai-Wah,² Marc W Deyell ²

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¹Division of Cardiology, Department of Medicine, University of British Columbia, Vancouver, British Columbia, Canada

²Heart Rhythm Services, Division of Cardiology, Department of Medicine, University of British Columbia, Vancouver, British Columbia, Canada

Correspondence to

Dr Marc W Deyell, Heart Rhythm Services, Division of Cardiology, Department of Medicine, University of British Columbia, Vancouver, BC V6T 1Z1, Canada; mdeyell@mail.ubc.ca

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ABSTRACT

Objective Despite the widespread and increasing use of ambulatory electrocardiography (ECG), there is no consensus on reference ranges for ambulatory electrocardiogram parameters to guide interpretation. We sought to determine population distribution-based reference ranges for parameters measured during ambulatory electrocardiogram in healthy adults, based on existing literature.

Methods We searched multiple databases from 1950 to 2020. Articles reporting original data from ≥24-hour ambulatory electrocardiogram monitoring in healthy adults were included. Data extraction and synthesis were performed according to Meta-analysis of Observational Studies in Epidemiology guidelines. The prevalence/mean and SD for common parameters (sinus pauses, conduction abnormalities and ectopy) were extracted by age group (18–39, 40–59, 60–79 and 80+ years).

Results We identified 33 studies involving 6466 patients. Sinus pauses of >3 s were rare (pooled prevalence <1%) across all ages. Supraventricular ectopy of >1000/24 hours increased with age, from 0% (95% CI 0% to 0%) in those aged 18–39 years to 6% (95% CI 0% to 17%) in those aged 60–79 years. Episodes of supraventricular tachycardia increased from 3% (95% CI 1% to 6%) in those aged 18–39 years to 28% (95% CI 9% to 52%) in those aged 60–79 years. Ventricular ectopy of >1000/24 hours also increased with age, from 1% (95% CI 0% to 2%) in those aged 18–39 years to 5% (95% CI 1% to 10%) in those aged 60–79 years. Episodes of non-sustained ventricular tachycardia ranged from 0% (95% CI 0% to 1%) in those aged 18–39 years to 2% (95% CI 0% to 5%) in those aged 60–79 years.

Conclusion Despite the limitations of existing published data, this meta-analysis provides evidence-based reference ranges for ambulatory electrocardiogram parameters and highlights significant age-dependent differences that should be taken into account during interpretation.

INTRODUCTION

Ambulatory electrocardiography or Holter monitoring is frequently used by primary care and specialist physicians for both diagnostic and monitoring purposes.^{1–3} Interpretation involves correlating patient symptoms with recorded arrhythmias, as well as the identification of asymptomatic arrhythmias. However, determining the significance of findings requires knowledge of

what variations of heart rhythm fall within the range of normal physiology. Reference ranges for quantitative diagnostic tests are most commonly initially based on the underlying population distribution, with those values falling outside the central 95% or 99% of the population being considered ‘abnormal’.⁴ The standard for determining such a reference range or value is to derive it from a sufficient sample of healthy subjects.⁵ Establishing reference ranges based on population distributions is a fundamental first step for diagnostic testing.

Despite extensive literature addressing reference ranges for standard resting electrocardiogram parameters,^{6–9} guidelines and consensus statements regarding ambulatory electrocardiogram monitoring offer no guidance on reference ranges for commonly measured parameters.^{10 11} Ambulatory electrocardiogram monitoring is dramatically increasing, particularly with the advent of novel monitoring technologies.¹² As such, there is a need to establish evidence-based reference ranges for ambulatory electrocardiogram parameters to guide interpretation and clinical care.

The purpose of this systematic review and meta-analysis was to determine population distribution-based reference ranges for commonly measured ambulatory electrocardiogram parameters among healthy adults.

METHODS

This systematic review and meta-analysis was performed and reported in accordance with the Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines.¹³ The MOOSE checklist is provided in online supplementary material, eTable 1. Patients and the public were not involved in the design or conduct of this research. As this study consisted only of a systematic review and meta-analysis, approval from our institutional ethics committee was not undertaken.

Search strategy

Relevant articles were identified using search strategies for MEDLINE, Cochrane CENTRAL, CINAHL, Embase, Web of Science and Clinical-Trials.gov from 1950 to April 2020, using the OVID software. Medical Subject Headings (MeSH) terms and relevant synonyms combining ‘healthy adults’ with ‘ambulatory electrocardiography’ and ‘arrhythmias, cardiac’ were used to identify relevant articles. MeSH terms identifying known

cardiac diseases or antiarrhythmic drugs were used to exclude any studies not limited to healthy participants. The complete search strategy, developed by the investigators, is provided in the online supplementary material, eTable 2. Reference lists of selected studies and review studies were also manually screened for additional studies. We did not identify any abstracts or references to unpublished data.

Studies meeting the following criteria were included: (1) ≥ 24 hour ambulatory electrocardiogram monitoring of healthy adults aged ≥ 18 years, (2) a minimum sample of 20 patients and (3) reporting of original data. We excluded studies that explicitly included patients with hypertension, diabetes and dyslipidaemia as each carries a high rate of cardiovascular disease. We also excluded studies focused on endurance athletes as these would not be generalisable to the general population.

Studies were included if they reported prevalence or mean values for any of the following parameters:

- ▶ Conduction abnormalities/pauses: sinus pauses, second-degree atrioventricular (AV) block type I.
- ▶ Supraventricular ectopy (SVE): presence of any SVE, frequency of SVE and presence of any supraventricular tachycardia (SVT) of ≥ 100 beats/min for ≥ 3 beats.
- ▶ Ventricular ectopy (VE): presence of any VE, frequency of VE and presence of any non-sustained ventricular tachycardia (VT) of ≥ 100 beats/min for ≥ 3 beats.
- ▶ Mean, minimum and maximum heart rate.

Those ambulatory electrocardiogram parameters considered abnormal at any level (such as atrial fibrillation, type II second-degree AV block, third-degree AV block) were not evaluated.

Abstracts were reviewed by one of two investigators (CW and MWD), and full-text review was completed independently by both. Where it was not evident if patients with hypertension, diabetes or dyslipidaemia were included, the characteristics of the study population were examined and the decision to include the study was adjudicated by agreement. To avoid multiple publication bias in serial reporting of the same cohort, the larger or more recent of the series was included. All non-English studies were translated and included.

To assess for bias, the Joanna Briggs Institute risk of bias analysis tool was used, which assesses nine parameters.¹⁴ Studies were categorised as having a low, moderate, high or critical risk of bias.

Data extraction

Information extracted from each study included population characteristics, sampling method, extent of screening for cardiac disease in addition to the prevalence or mean values for the ambulatory electrocardiogram parameters outlined previously. Data were extracted independently by two investigators (CW and MWD). Results were extracted for the overall study cohort and, when available, by the following prespecified age strata: 18–39, 40–59, 60–79 and 80+ years. These age intervals were chosen as pragmatic to simplify recommendations for broad age groups. For articles that did not divide their cohorts into these specific age brackets, mean age and age range were used to categorise the data into the most appropriate cohort, where possible. Ectopic beat frequency was converted into total number in 24 hours.

Statistics

We prespecified age-stratified analysis of all parameters. Proportions with exact binomial 95% CIs were determined for each dichotomous ambulatory electrocardiogram variable.

For the meta-analysis, pooling of dichotomous parameters was performed using a random effects model with arcsine transformation of the proportions to stabilise the variances.¹⁵ Overall and within-age group heterogeneities were assessed using the I^2 measure.

Meta-analysis of continuous variables was performed by calculating the 95% CIs around the mean value, from the SD, assuming a normal distribution. Weighted averages of the mean value, lower 95% confidence bound and upper 95% CI bound were calculated.

Publication bias for each parameter was assessed by examination of funnel plots. Funnel plots for the proportional meta-analyses were performed by plotting study sample size over log odds.¹⁶ All analyses were performed using STATA V.13.1.

RESULTS

Study selection

This search retrieved 926 articles. After abstract review, 847 articles were unrelated to the study objectives, leaving 79 articles for full-text review. Six additional studies were identified from scanning reference lists (online supplementary eFigure 1). Thirty-three studies involving 6466 patients met the inclusion and exclusion criteria. Details regarding the study population, methods of screening for underlying cardiovascular disease and sampling method are provided in table 1.

The majority of studies were published prior to 1990, with only five published after 2000. Sample sizes ranged from 26 to 2048, with a median of 86. All were prospective except for the Hingorani *et al*¹⁷ and Mikulecký and Kujaník¹⁸ studies, with the majority using convenience sampling. The most common methods for disease screening included history, physical and screening electrocardiogram. Seven studies performed echocardiography (624 patients, 10%); four studies (337 patients, 5%) performed exercise stress testing; and two studies (271 patients, 4%) included coronary angiograms. Six studies reported smoking status without quantifying amount. No studies allowed the use of any cardiac medications.

Thirteen studies exhibited moderate risk of bias, mostly secondary to small sample sizes and limited disease screening. The remainder had low risk of bias, and none were high or critical risk (table 1).

Sinus pauses and conduction abnormalities

Eight and five studies reported the proportion of patients with sinus pauses of >2 and >3 s, respectively. Data on circadian variation in sinus pauses (nocturnal vs daytime) were not reported in most published studies and were not consistently reported in the remainder to allow pooling. No studies reported the proportion of sinus pauses in those aged 80 years and older. Sinus pauses of >2 s were relatively common with a pooled estimate of 3% (95% CI 0% to 7%) and showed a U-shaped distribution with the highest proportion in those aged <40 years and again increasing in those aged 60 years and over (table 2 and online supplementary eFigure 2A). Sinus pauses of >3 s (table 2 and online supplementary eFigure 2B) were not observed in any of the six studies, leading to a pooled estimate of 0% (95% CI 0% to 0%).

Ten studies reported on the proportion of patients with any first-degree AV block (table 2 and online supplementary eFigure 3). The pooled estimate for first-degree AV block was 3% (95% CI 1% to 6%). Only the age group of 40–59 years had a pooled estimate of $\leq 1\%$ or less. The highest rates were seen in those aged 18–39 years; however, there was significant heterogeneity in this group. Only one study reported on those aged 80+ years.

Arrhythmias and sudden death

Table 1 Summary of included studies

First author	Year	N	Study population	Prospective	Convenience sampling	Screening	Risk of bias
Clarke <i>et al</i> ³¹	1976	86	Volunteers from a company	Y	Y	Hx, Px, electrocardiogram, Bw	Low
Raftery ³²	1976	55	Patients from a general practice	Y	Y	Hx, Px, electrocardiogram	Low
Brotsky ³³	1977	50	Recruited male medical students	Y	Y	Hx, Px, CXR, electrocardiogram, Echo	Low
Adey ³⁴	1978	189	Recruited from local service clubs, industrial organisations and the hospital	Y	Y	Hx, Px, CXR, electrocardiogram— included smokers	Moderate
Kostis ³⁵	1981	101	Subgroup of patients referred for cardiac catheterisation	Y	Y	Hx, Px, Bw, electrocardiogram, Echo, Coronary angiogram, stress test	Low
Sobotka ³⁶	1981	67	Students/graduates of healthcare professional fields	Y	Y	Hx, Px, CXR, electrocardiogram and Echo	Low
Bjerregaard ³⁷	1982	260	Members of a leisure club and jogging club	Y	Y	Hx, Px, CXR, electrocardiogram	Low
Bethege ³⁸	1983	170	Colleagues, families, friends and non-hospitalised patients	Y	Y	Hx, Px, electrocardiogram. If >40, stress test±Echo±angiograms	Low
Romhilt ³⁹	1984	101	Female employees from a company, randomly selected	Y	Y	Hx, Px, CXR, electrocardiogram	Low
Liu ⁴⁰	1985	130	Clinic patients	Y	Y	Hx, Px, Bw, CXR, electrocardiogram	Low
Matsuno ⁴¹	1985	94	Clinic patients, at annual physical exams	Y	Y	Hx, Px, electrocardiogram, Bw	Low
Rasmussen ⁴²	1985	111	Randomly selected adults from Copenhagen City Heart Study	Y	Y	Hx, Px, electrocardiogram, Bw	Low
Kantelip ⁴³	1986	50	Healthy patients—recruitment unclear	Y	Y	Hx, Px, Bw, electrocardiogram	Moderate
Grosogoeat ⁴⁴	1986	134	Clinic patients, at annual physical exams	Y	Y	Hx, Px, Bw, electrocardiogram, CXR	Low
Takeuchi ⁴⁵	1986	46	Healthy patients—recruitment unclear	Y	Y	Hx, Px, electrocardiogram	Moderate
Dietz ⁴⁶	1987	50	Volunteers <60 years of age	Y	Y	Hx, Px, electrocardiogram	Low
Mammarella ⁴⁷	1987	68	Healthy patients—recruitment unclear	Y	Y	Hx	Moderate
Andersson ⁴⁸	1988	32	Elderly from a single town in Sweden	Y	Y	Hx, Px	Moderate
Neto ⁴⁹	1988	50	Clinic patients >65 years of age	Y	Y	Hx, Px	Low
Suwa ⁵⁰	1988	179	Volunteers	Y	Y	Hx, Px, electrocardiogram	Low
Ribera ⁵¹	1989	50	Retired hospital workers	Y	Y	Hx, Px, electrocardiogram	Moderate
Takada ⁵²	1989	220	Clinic patients	Y	Y	Hx, Px, electrocardiogram, CXR, Bw	Low
Manyan ⁵³	1990	86	Healthy volunteers (unknown how recruited)	Y	Y	Hx, Px, Bw, CXR, electrocardiogram, Echo, spirometry	Moderate
Wajngarten ⁵⁴	1990	26	Men from a veterans association	Y	Y	Hx, Px, Bw, CXR, Echo, Exercise electrocardiogram, Gated Cardiac Imaging	Moderate
Bjerregaard ⁵⁵	1991	260	Individual from a senior leisure club, jogging club and their relatives	Y	Y	Hx, Px, electrocardiogram, CXR	Low
Buerschaper ⁵⁶	1991	79	Healthy family members and acquaintances of clinic patients	Y	Y	Hx, Px, electrocardiogram, treadmill test	Low
Palatini ⁵⁷	1992	40	Men from a sports medicine centre	Y	Y	Hx, Px, Bw, electrocardiogram, Echo, stress test	Low
Wakida ⁵⁸	1994	46	All centenarians in Aichi, Japan	Y	N	Hx, Px, electrocardiogram	Moderate
Mikulecky ¹⁸	2008	37	Random healthy men from Hungary with electrocardiograms on file	N	Y	Hx, Px, electrocardiogram	Moderate
Hingorani ¹⁷	2016	1273	Participants of 22 phase I clinical trials	N	N	Hx, Px, Bw, electrocardiogram, normal BMI	Low
Von Rotz ⁵⁹	2017	2048	All inhabitants of Liechtenstein, aged 25–41 years	Y	N	Hx, Px, Bw—excluded high BMI	Low
Zorzi (1) ⁶⁰	2018	144	Sedentary controls, volunteers, for study of athletes, aged 16–35 years	Y	Y	Hx	Moderate
Zorz (2) ⁶¹	2018	134	Sedentary controls, volunteers for study of athletes, aged 39–51 years	Y	Y	Hx	Moderate

Please refer to the online supplementary material for the reference details of references 31–60.

BMI, body mass index; Bw, bloodwork; CXR, chest x-ray; Echo, echocardiogram; Hx, history; N, no; Px, physical examination; Y, yes.

Twelve studies reported the proportion of patients with second-degree AV block, type I. The pooled estimate was 2% (95% CI 1% to 2%). There was no heterogeneity between studies or between age groups (table 2 and online supplementary eFigure 4).

Interpretation of the funnel plots for the conduction system disease parameters was limited by the small number of studies, but there was asymmetry (online supplementary eFigure 5).

Supraventricular ectopy and tachycardia

Twenty-one studies reported on the presence of any SVE. The prevalence of any SVE in a 24-hour period was high, with a pooled estimate of 64% (95% CI 52% to 76%). There was significant

heterogeneity in the prevalence of SVE between studies in the same age strata that was not explained by differences in study design. The pooled estimate for presence of SVEs increased significantly with age, from 43% in 18–39 years to 80% in 60–79 years (table 3 and online supplementary eFigure 6).

Few studies reported the frequency of SVE. Only three studies reported the proportion of patients having >500 SVE in 24 hours. Having >500 SVE in 24 hours was rare in those <40 years of age with a pooled estimate of 0% (95% CI 0% to 1%), but the prevalence rose significantly in older age groups (table 3 and figure 1A). Only three studies reported the proportion of patients with >1000 SVE/24 hours, and similar findings were

Table 2 Summary of prevalence of sinus pauses and conduction block on ambulatory electrocardiogram in healthy populations

Age group (years)	Sinus pauses ≥ 2 s		Sinus pauses ≥ 3 s		First-degree AV block		Second-degree AV Block mobitz type 1	
	N	% (95% CI)	N	% (95% CI)	N	% (95% CI)	N	% (95% CI)
18–39	1400	7 (1 to 15)	1282	0 (0 to 0)	1372	6 (1 to 15)	1400	2 (1 to 3)
40–59	178	0 (0 to 1)	108	0 (0 to 3)	453	1 (0 to 2)	453	1 (0 to 2)
60–79	63	1 (0 to 6)	37	0 (0 to 9)	91	2 (0 to 6)	123	1 (0 to 5)
80+	–	–	–	–	21	19 (5 to 42)	71	1 (0 to 6)
All	1641	3 (0 to 7)	1427	0 (0 to 0)	1958	3 (1 to 6)	2047	2 (1 to 2)

The weighted, pooled prevalence and 95% CIs from a random effects model are presented. N represents the total population size from which the pooled estimate was derived. AV, atrioventricular.

observed (table 3 and figure 1B). There was significant heterogeneity between age groups for all cut-offs of SVE frequency that was mostly related to an increase in SVE with age.

Seventeen studies reported on the presence of any SVT, defined by the presence of 3–5 consecutive ectopic atrial beats (the minimum duration differed between studies). The overall pooled estimate was 11% (95% CI 4% to 19%). Again, the proportion of patients with SVT significantly increased with age, reaching 33% (95% CI 0% to 82%) of patients aged 60–79 years (table 3 and figure 2). There was insufficient data on the number and length of SVT episodes to determine either an upper limit of number of episodes or a maximum allowable duration that would be considered within the normal range.

There was asymmetry in the funnel plots for >500 and >1000 SVEs in 24 hours, driven largely by a single large study in younger individuals showing a low rate of ectopy (online supplementary eFigure 7).

Ventricular ectopy and tachycardia

Thirty-one studies reported on the presence of any VE (table 3 and online supplementary eFigure 8). Having any VE was common with a pooled estimate of 55% (95% CI 49% to 61%). Similar to SVE, the proportion of patients with any VE significantly increased with age, ranging from 40% in those aged 18–39 years to 84% in those aged 80+ years. There was significant heterogeneity within each age group.

Considerably more studies reported VE than SVE frequency. Ten studies reported on the proportion of patients with >500 VE/24 hours (table 3 and figure 3A). Having >500 VE/24 hours was rare in those aged 18–39 years (pooled estimate 1%, 95% CI

0% to 3%). The proportion rose in the older age groups. Similarly, 10 studies reported on the proportion of patients with >1000 VE/24 hours (table 3 and figure 3B). Having >1000 VE/24 hours was uncommon (pooled estimate $\leq 1\%$) in those aged 18–39 and 40–69 years but increased with age.

Seventeen studies reported on the presence of any non-sustained VT (table 3 and figure 4). Non-sustained VT was defined by the presence of ≥ 3 consecutive ventricular beats (the minimum duration differed between studies from 3 to 5 beats). The overall pooled estimate was 1% (95% CI 0% to 2%). Again, there was a slight increase in the proportion of patients with VT in increasing age groups with an estimate of 2% of patients aged 60–79 years. There were insufficient data on length of VT episodes to determine a maximum allowable duration that would be considered within the normal range.

Asymmetry in the funnel plots for non-sustained VT and VE burden was caused by a single large study in younger subjects who had a low proportion of these findings (online supplementary eFigure 9).

Heart rate

The meta-analysis for minimum, mean and maximum heart rates is found in the online supplementary material, eFigures 10 and 11.

DISCUSSION

This is the first systematic review and meta-analysis of population distribution-based reference ranges for commonly measured electrocardiogram parameters. The foremost finding of this systematic review is that age must be taken into account

Table 3 Summary of prevalence of ectopy and tachycardia on ambulatory electrocardiogram in healthy populations

Age group (years)	Any SVE		SVE>500 in 24 hours		SVE>1000 in 24 hours		Any SVT	
	N	% (95% CI)	N	% (95% CI)	N	% (95% CI)	N	% (95% CI)
18–39	1508	43 (29 to 58)	1293	0 (0 to 1)	1293	0 (0 to 1)	1377	5 (1 to 12)
40–59	850	56 (28 to 32)	18	6 (0 to 27)	18	0 (0 to 19)	676	6 (2 to 13)
60–79	374	80 (57 to 96)	38	16 (5 to 30)	38	6 (0 to 17)	489	33 (0 to 82)
80–99	93	90 (47 to 100)	22	27 (11 to 50)	22	27 (11 to 50)	50	10 (3 to 22)
All	2825	64 (52 to 76)	1371	7 (0 to 21)	1371	3 (0 to 13)	2592	11 (4,19)

Age group (years)	Any VE		VE>500 in 24 hours		VE>1000 in 24 hours		Any NSVT	
	N	% (95% CI)	N	% (95% CI)	N	% (95% CI)	N	% (95% CI)
18–39	4076	41 (30 to 53)	3377	1 (0 to 3)	3377	1 (0 to 2)	1534	0 (0 to 1)
40–59	1805	53 (43 to 62)	692	2 (1 to 3)	326	1 (0 to 3)	992	1 (0 to 2)
60–79	429	70 (61 to 79)	75	6 (0 to 18)	233	5 (1 to 10)	214	2 (0 to 5)
80–99	93	84 (54 to 100)	22	5 (0 to 23)	72	5 (1 to 12)	71	2 (0 to 8)
All	6125	56 (49 to 62)	4166	2 (1 to 3)	4008	2 (1 to 3)	2533	1 (0 to 2)

The weighted, pooled prevalence and 95% CIs from a random effects model are presented. N represents the total population size from which the pooled estimate was derived. NSVT, non-sustained ventricular tachycardia; SVE, supraventricular ectopy; SVT, supraventricular tachycardia; VE, ventricular ectopy.

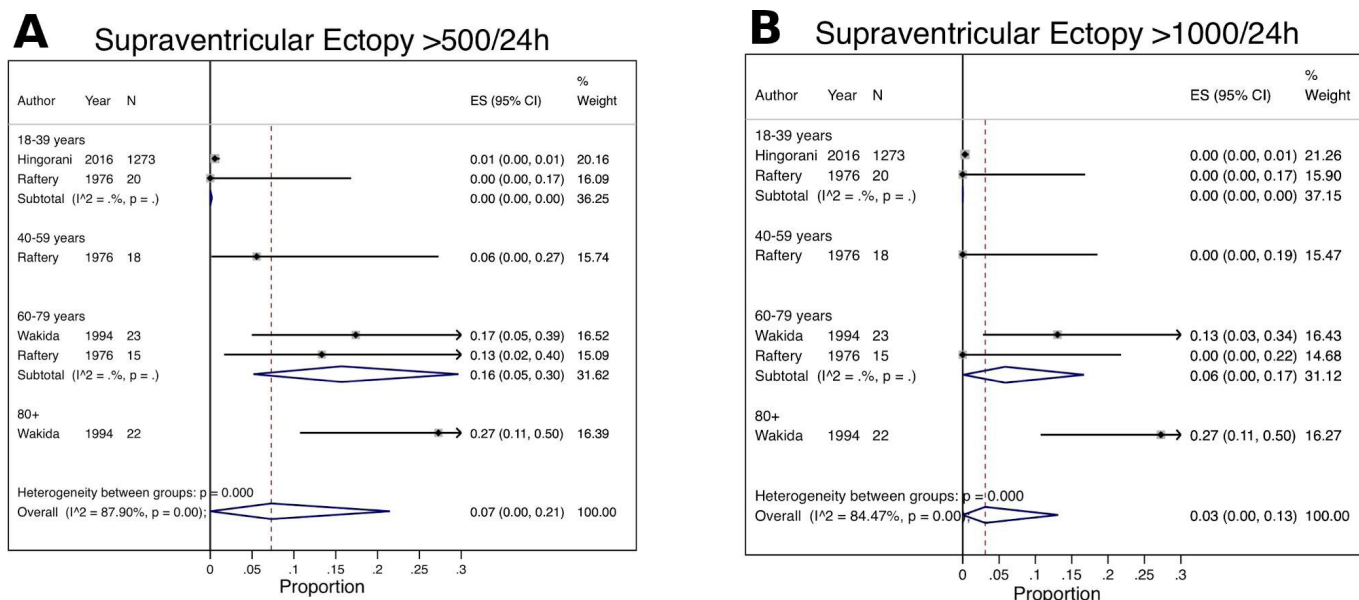


Figure 1 Forest plots for the prevalence of >500 and >1000 supraventricular ectopic beats in 24 hours ES denote the weighted mean prevalence with >500 SVE (A) and >1000 SVE (B) in 24 hours, by age group and overall. Note that within-age group heterogeneity could not be tested for some groups due to the small number of studies and/or no events in some studies. ES, effect size; SVE, supraventricular ectopy.

for ambulatory electrocardiogram interpretations due to age-dependent variation in many parameters. This observation was particularly of relevance for the determination of what might constitute an abnormal frequency of SVE, SVT, VE and non-sustained VT. Another key finding was the overall paucity of high-quality data on reference ranges for ambulatory electrocardiogram parameters, particularly for patients aged >40 years.

An important feature of this systematic review is the inclusion of truly healthy subjects, excluding those with disease and cardiac risk factors. Determining reference ranges from healthy population distributions is the most widely accepted methodology for the initial establishment of reference ranges.⁵ Three sizeable studies of ambulatory electrocardiogram data in 'healthy individuals' that were not included¹⁹⁻²² because the cohorts had significant rates of hypertension, diabetes, smoking, prescription of cardiac medications or known stable coronary artery disease. When comparing this systematic review to the large Copenhagen study,²⁰ the latter found a higher proportion of participants with any VE and excessive VE. A similar trend was found in the large Cardiovascular Health Study.¹⁹ Whether the increased frequency of VE in these studies is attributable to the higher prevalence of cardiac risk factors or overt coronary disease is unclear. Thus, while these studies are important contributions, their inclusion in the current meta-analysis would add bias towards overestimating the reference ranges.

We recognise that the ultimate desire for clinicians is not to necessarily know the population distribution-based reference ranges but rather cut-offs based on prognosis or treatment, termed 'medical decision limits'. While desirable, establishing such limits is very difficult. Observational studies can rarely provide robust medical decision limits, given the vulnerability of their estimates from bias introduced with their design. In the case of ambulatory electrocardiogram parameters, it would be nearly impossible to determine a robust association between a parameter and outcomes, given these parameters may change considerably over time.

This is highlighted by recent literature examining outcomes with excessive SVE and VE. Excessive SVE has been correlated with an increased risk of atrial fibrillation (HR ranging from 2.7 to 3.2)^{21 23} and stroke (HR 1.9-2.79).^{21 24} However, the definition of excessive atrial activity across studies is inconsistent, ranging from three per hour (100 per 24 hours) to 30 per hour (720 per 24 hours), with suggestions that even 100 per 24 hours may be clinically significant.²³ None of the studies accounted for changes in SVE frequency over time, making the association

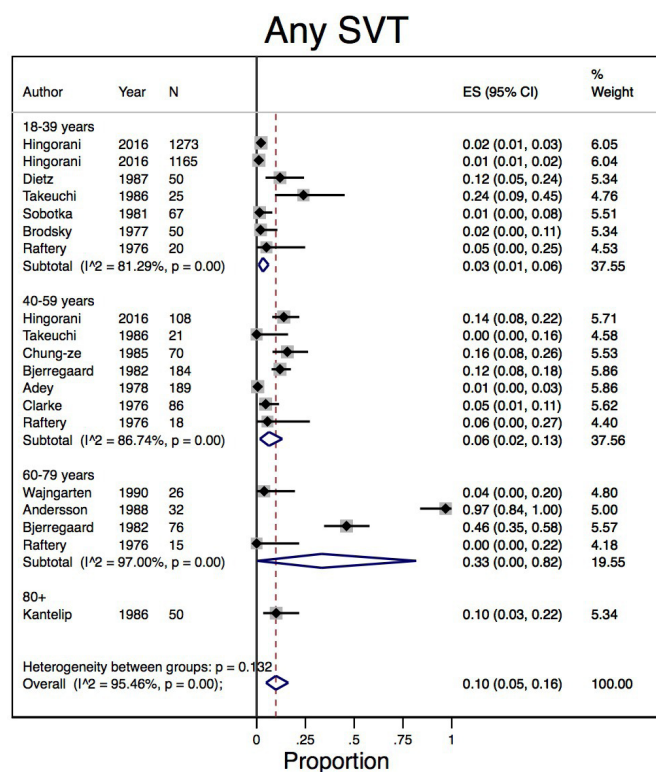


Figure 2 Forest plot for the prevalence of any SVT ES denotes the weighted mean prevalence with any SVT, by age group and overall. ES, effect size; SVT, supraventricular tachycardia.

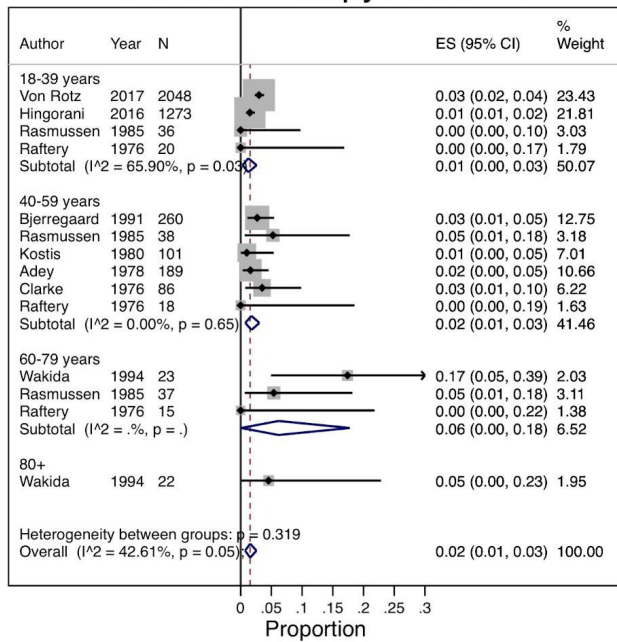
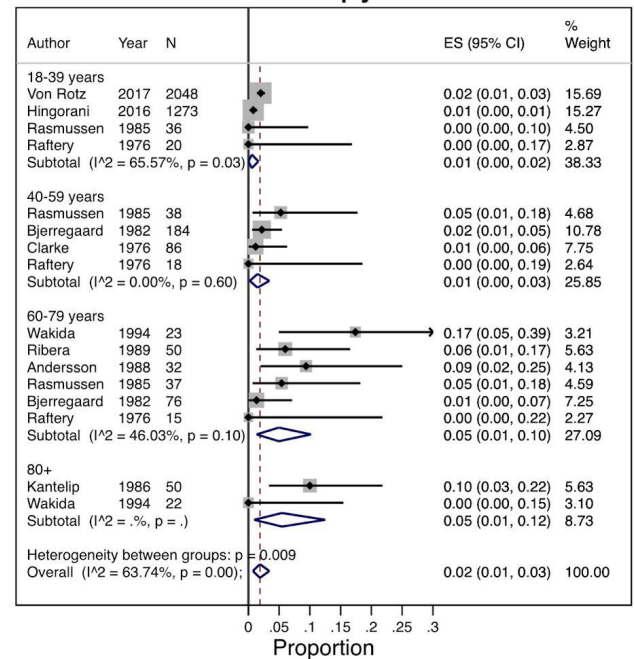
A Ventricular Ectopy >500/24h**B** Ventricular Ectopy >1000/24h

Figure 3 Forest plots for the prevalence of >1000 and >500 ventricular ectopic beats in 24 hours ES denote the weighted mean prevalence of >500 VE (A) and >1000 VE (B) in 24 hours, by age group and overall. Note that within-age group heterogeneity could not be tested for some groups due to the small number of studies and/or no events in some studies. ES, effect size; VE, ventricular ectopy.

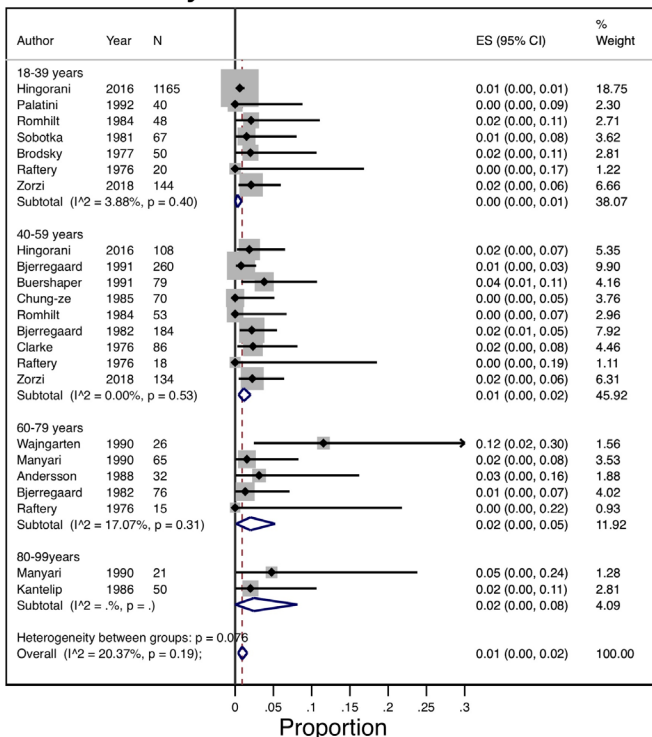
Any Non-Sustained VT

Figure 4 Forest plot for the prevalence with any non-sustained VT ES denotes the weighted mean prevalence with any non-sustained VT, by age group and overall. Note that within-age group heterogeneity could not be tested for some groups due to the small number of studies and/or no events in some studies. ES, effect size; VT, ventricular tachycardia.

between the measurement and outcomes dubious. Furthermore, based on the findings of the current study, a large proportion of apparently healthy participants would be categorised 'at risk'.

Excessive VEs, similarly, are correlated with adverse outcomes, including increased risk of incident heart failure (HR 1.48)¹⁹ and sudden cardiac death (HR 1.31–3.0).^{6 19 20 22 25–28} However, the amount of excessive VE required to confer increased risk varied widely, ranging from 1 per hour to 360 per hour.²⁰ No studies accounted for changes in VE burden over time, which can be considerable.²⁹ Greater than 1000 VE/24 hours may be a normal finding based on population distribution, particularly in older age groups.

Establishing robust medical decision limits (prognostic cut-offs) for ambulatory electrocardiogram parameters will only come with randomised trials of intervention (treatment or enhanced monitoring) based on parameter thresholds. This is how other medical decision limits, such as those for dyslipidaemia, were established and continue to evolve.³⁰ As we await such trials, population distribution-based reference ranges will remain the cornerstone for ambulatory electrocardiogram interpretation.

Limitations

The key limitations of this systematic review relate to the existing literature. First, the screening methods for identifying truly healthy subjects varied between studies, and it is possible that patients with subclinical heart disease were included. Second, there was a paucity or absence of data for certain ambulatory electrocardiogram parameters. Only two studies had a sample size of >1000 patients. There were minimal data on SVE frequency in those over 40 years.

There were insufficient available data on the frequency, duration or rate of SVT or non-sustained VT to determine a threshold for what may be considered abnormal. Also, the data regarding

Table 4 Suggested recommendations for reporting of abnormal findings on ambulatory electrocardiogram monitoring

Age group (years)	Sinus pauses	1° AV block	2° AV block, type I	SVE frequency	Any SVT	VE frequency	Any non-sustained VT
18–39	Abnormal: >3 s	May be normal finding	May be normal finding	Normal: 0–500/24 hours Abnormal: >500/24 hours	May be a normal finding	Normal: 0–500/24 hours Abnormal: >500/24 hours	Abnormal
40–59	Abnormal: >2 s	Abnormal	Abnormal	Normal: 0–1000/24 hours Abnormal: >1000/24 hours	May be a normal finding	Normal: 0–1000/24 hours Abnormal: >1000/24 hours	Abnormal
60–79	Abnormal: >3 s	May be normal finding	Abnormal	Not available (>1000/24 hours may be a normal finding)	May be a normal finding	Not available (>1000/24 hours may be a normal finding)	May be a normal finding
80+	Not available	May be normal finding	Abnormal	Not available (>1000/24 hours may be a normal finding)	May be a normal finding	Not available (>1000/24 hours may be a normal finding)	May be a normal finding

The above recommendations are based on the population distribution for a given parameter. Any parameter present in $\leq 1\%$ of the population was considered abnormal. AV, atrioventricular; SVE, supraventricular ectopy; SVT, supraventricular tachycardia; VE, ventricular ectopy; VT, ventricular tachycardia.

prevalence and duration of sinus pauses were sparse. There were also insufficient data to conduct a sex-specific analysis of reference ranges.

Recommendations for ambulatory electrocardiogram reporting

This systematic review provides evidence-based direction for what should pragmatically be considered an abnormal finding on an ambulatory electrocardiogram study. Given the limitations of the existing literature, we conservatively defined an abnormal finding as being present in $\leq 1\%$ of the population (as opposed to a more standard $\leq 5\%$) for dichotomous variables. Practical suggestions for defining abnormal ambulatory electrocardiogram findings are proposed in table 4. It must be emphasised, however, that these are suggested reporting guidelines based on the available data in the literature and do not replace the full synthesis of the ambulatory electrocardiogram data within the clinical context in which the test was performed.

Key messages

What is already known on this subject?

- Despite the widespread use of ambulatory electrocardiogram monitoring in clinical practice, there are no current guidelines with respect to interpretation of ambulatory electrocardiogram studies or what normal ranges are for commonly measured parameters. To date, only one non-systematic review has examined reference ranges for ambulatory electrocardiogram parameters.

What might this study add?

- This meta-analysis summarises existing data on reference ranges for ambulatory electrocardiogram parameters. Despite the limitations of the existing data, the current study provides evidence-based estimates of the population distribution of common ambulatory electrocardiogram parameters. It also identified significant age-dependent changes in reference ranges.

How might this impact on clinical practice?

- The results of this meta-analysis will allow for more evidence-based interpretation of ambulatory electrocardiogram studies to help guide not only interpretation of the studies but also subsequent care.

CONCLUSION

Despite the limitations of existing data, this meta-analysis provides evidence-based estimates of reference ranges for common ambulatory electrocardiogram parameters based on population distributions. There are important age-related differences for many parameters that should be taken into consideration during interpretation of ambulatory electrocardiogram studies.

Twitter Marc W Deyell @MarcDeyell

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ORCID iD

Marc W Deyell <http://orcid.org/0000-0001-8434-7076>

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